

D414

Cyber Operations Fundamentals

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

An online retailer, which has 1.5 million loyalty members, finds that the prices in its product database were changed without authorization overnight. No data left the network, and the site stayed up. Which part of the CIA triad was violated?

- A. All three, since any incident breaks all three.
- B. Confidentiality.
- C. Availability.
- D. Integrity.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A flood of traffic makes a hospital's patient portal unreachable for six hours. Which CIA property is most directly affected?

- A. Non-repudiation.
- B. Integrity.
- C. Availability.
- D. Confidentiality.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A SOC analyst at a utility company, whose operations center has large wall displays, must find out exactly what file was sent in a suspicious transfer. Only NetFlow records are available for that time. What can she determine?

- A. The user's password.
- B. The hosts, ports, time, and bytes transferred — but not the file contents.
- C. The full file contents.
- D. Nothing, since NetFlow records no useful data.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

An analyst needs to reconstruct the exact commands sent in an unencrypted session. Which data source is required?

- A.** Full packet capture.
- B.** DHCP logs.
- C.** NetFlow.
- D.** Firewall allow/deny logs.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A firewall engineer, who has worked the night shift for three years, must write a rule to block one suspicious TCP connection. Which set of fields identifies the connection?

- A.** Source IP, destination IP, source port, destination port, protocol.
- B.** Source IP and destination IP only.
- C.** Username, time, and URL.
- D.** Source MAC, destination MAC, VLAN, hostname, protocol.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

An email with a malicious attachment reaches employees at a shipping company that recently won a safety award. Nobody has opened it yet. Which Kill Chain stage has the attacker reached?

- A.** Reconnaissance.
- B.** Exploitation.
- C.** Command and control.
- D.** Delivery.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

Defenders block outbound connections from an infected host to the attacker's server. Which stage are they disrupting?

- A.** Weaponization.
- B.** It is too late; no stage can be disrupted after infection.
- C.** Delivery.
- D.** Command and control.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A bank's network IDS, installed by a respected integrator, sees HTTPS traffic from a workstation to an unknown site but cannot see what is inside. Which tool would best reveal what the workstation is doing?

- A. The integrator's reputation.
- B. A host-based agent on the workstation.
- C. A second network IDS.
- D. A faster network link.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A defense contractor, located near an Air Force base, gives each document a label like Secret or Top Secret. Users can only open documents at or below their clearance, and owners cannot change this. Which access control model is this?

- A. Role-based access control (RBAC).
- B. Discretionary access control (DAC).
- C. Mandatory access control (MAC).
- D. Any model; they are all the same.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A hospital grants chart access based on job title: nurses, physicians, and billing staff each get a set of permissions. Which model is this?

- A. RBAC.
- B. ABAC, because roles are attributes.
- C. DAC.
- D. MAC.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A school district's web server, which hosts the lunch menu, runs software with a known unpatched flaw. What is the flaw itself?

- A.** A threat.
- B.** A vulnerability.
- C.** A risk.
- D.** An exploit.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

An IDS at a university, which just started its fall semester, alerts on a login from a foreign country. Investigation shows it was a professor traveling for a conference, using her own valid account. How is the alert classified?

- A.** True negative.
- B.** False negative.
- C.** False positive.
- D.** True positive.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

An online retailer, ~~which has 1.5 million loyalty members~~, finds that the prices in its product database were changed without authorization overnight. No data left the network, and the site stayed up. Which part of the CIA triad was violated?

Target word: CIA property **Decides:** Data was altered.

A	Absolute language	Only integrity was harmed here.
B	Confusion set	No data was exposed.
C	Adjacent category	The site stayed up.
D	Correct	Unauthorized changes break integrity.

2. Answer C Concept 1 • transfer

A flood of traffic makes a hospital's patient portal unreachable for six hours. Which CIA property is most directly affected?

Target word: CIA property **Decides:** Users cannot access the service.

A	Adjacent category	About proof of actions, not access.
B	Confusion set	No data was changed.
C	Correct	The service could not be reached.
D	Inverted operation	Nothing was disclosed.

3. Answer B Concept 2

A SOC analyst at a utility company, ~~whose operations center has large wall displays~~, must find out exactly what file was sent in a suspicious transfer. Only NetFlow records are available for that time. What can she determine?

Target word: what NetFlow shows **Decides:** NetFlow is metadata only.

A	Adjacent category	Flow records hold no credentials or payloads.
B	Correct	Flow data records conversations, not payloads.
C	Confusion set	Only full packet capture holds contents.
D	Absolute language	Metadata is useful for scoping.

4. Answer A Concept 2 • transfer

An analyst needs to reconstruct the exact commands sent in an unencrypted session. Which data source is required?

Target word: data source for contents **Decides:** Needs full payload.

A	Correct	Holds complete packet contents.
B	Confusion set	Record address leases.
C	Inverted operation	Metadata only.
D	Adjacent category	Record decisions, not contents.

5. Answer A Concept 3

A firewall engineer, ~~who has worked the night shift for three years,~~ must write a rule to block one suspicious TCP connection. Which set of fields identifies the connection?

Target word: 5-tuple fields **Decides:** Standard flow identifier.

A	Correct	The 5-tuple.
B	Dropped element	Leaves out ports and protocol.
C	Adjacent category	Application-level fields.
D	Confusion set	Layer 2 and name fields, not the 5-tuple.

6. Answer D Concept 4

An email with a malicious attachment reaches employees at a shipping company ~~that recently won a safety award.~~ Nobody has opened it yet. Which Kill Chain stage has the attacker reached?

Target word: stage **Decides:** The payload has arrived but not run.

A	Confusion set	An earlier stage of research.
B	Inverted operation	Needs the code to run.
C	Adjacent category	A later stage after installation.
D	Correct	The weapon has reached the target.

7. Answer D Concept 4 · transfer

Defenders block outbound connections from an infected host to the attacker's server. Which stage are they disrupting?

Target word: stage disrupted **Decides:** Outbound callback to attacker infrastructure.

A	Confusion set	Happens on the attacker's side before delivery.
B	Absolute language	Breaking C2 still prevents actions on objectives.
C	Inverted operation	Delivery is inbound.
D	Correct	The channel the attacker uses to direct the host.

8. Answer B Concept 5

A bank's network IDS, ~~installed by a respected integrator,~~ sees HTTPS traffic from a workstation to an unknown site but cannot see what is inside. Which tool would best reveal what the workstation is doing?

Target word: better visibility **Decides:** Traffic is encrypted.

A	Camouflage	Not a visibility source.
B	Correct	Sees processes and activity before encryption.
C	Dropped element	Still cannot read encrypted traffic.
D	Adjacent category	Speed does not reveal contents.

9. Answer C Concept 6

A defense contractor, ~~located near an Air Force base~~, gives each document a label like Secret or Top Secret. Users can only open documents at or below their clearance, and owners cannot change this. Which access control model is this?

Target word: model **Decides:** System-enforced labels; owners cannot override.

A	Confusion set	Based on job roles, not labels.
B	Inverted operation	In DAC owners decide.
C	Correct	Policy labels decide access.
D	Absolute language	Models differ in who decides.

10. Answer A Concept 6 · transfer

A hospital grants chart access based on job title: nurses, physicians, and billing staff each get a set of permissions. Which model is this?

Target word: model **Decides:** Access by job role.

A	Correct	Permissions attach to roles.
B	Adjacent category	ABAC evaluates many attributes; this is role-only.
C	Inverted operation	Owners decide in DAC.
D	Confusion set	Uses classification labels.

11. Answer B Concept 7

A school district's web server, ~~which hosts the lunch menu~~, runs software with a known unpatched flaw. What is the flaw itself?

Target word: term **Decides:** A weakness.

A	Confusion set	A threat is the potential danger that might use it.
B	Correct	A weakness that could be used.
C	Adjacent category	Risk combines likelihood and impact.
D	Inverted operation	The means of using the flaw, not the flaw.

12. Answer C Concept 8

An IDS at a university, ~~which just started its fall semester~~, alerts on a login from a foreign country. Investigation shows it was a professor traveling for a conference, using her own valid account. How is the alert classified?

Target word: classification **Decides:** Alerted, but benign.

A	Adjacent category	Means no alert on benign activity.
B	Inverted operation	That is a missed attack.
C	Correct	An alert on legitimate activity.
D	Confusion set	Would mean a real attack.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	2	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	10	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	11	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	CIA triad	Confidentiality keeps data from unauthorized eyes; integrity keeps it accurate and unaltered; availability keeps it accessible when needed.	Calling a denial-of-service attack a confidentiality breach.	What was harmed: secrecy, accuracy, or access.
2	Security data types	Full packet capture holds complete contents; NetFlow and session data hold metadata (who talked to whom, when, how much); logs record events on devices.	Expecting NetFlow to show the contents of a file transfer.	Whether the analyst needs contents or only conversation details.
3	The 5-tuple	A connection is identified by source IP, destination IP, source port, destination port, and protocol.	Including MAC address or hostname in the 5-tuple.	Fields needed to identify or block a specific flow.
4	Cyber Kill Chain stages	Reconnaissance → weaponization → delivery → exploitation → installation → command and control → actions on objectives; defenders aim to break the chain as early as possible.	Mixing up delivery and exploitation, or believing only late stages can be stopped.	An observed attacker step and a question about its stage.
5	Host-based vs. network-based monitoring	Network sensors see traffic but not encrypted contents or local activity; host-based tools see processes, files, and decrypted activity on the endpoint.	Relying on a network IDS to inspect encrypted traffic.	Encrypted traffic or activity that never crosses the network.
6	Access control models	DAC lets owners grant access; MAC uses labels set by policy; RBAC grants rights by job role; ABAC evaluates attributes.	Calling any policy-based system RBAC.	Who decides access: owner, labels, role, or attributes.
7	Threat, vulnerability, and risk	A vulnerability is a weakness; a threat is a potential danger that could use it; risk is the likelihood and impact of that happening.	Calling an unpatched flaw a threat.	A weakness vs. an actor or event vs. a combined judgment.
8	Alert classification	True positive = real attack alerted; false positive = benign activity alerted; false negative = real attack missed; true negative = benign activity not alerted.	Swapping false positive and false negative.	An alert outcome after investigation.

Course scope (public description): In Cyber Operations Fundamentals, students will learn security concepts, security monitoring, host-based analysis, network intrusion analysis, and security policies and procedures using Cisco practices and technologies.